

Pattern: Each row in the dataset, (data point / Instance)

Feature: Each Column in the dataset

Classes: Types or Categories into which patterns are classified

Metric

Let $X \neq \emptyset$ be a set Dataset

Ex $x = \begin{bmatrix} 5 \\ 3.3 \\ 1.4 \\ 0.2 \end{bmatrix} \begin{matrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{matrix}$ $x \in \mathbb{R}^d$
 $x \in X$
 $\#$ is a sample
 4×1 i.e. $(d \times 1)$

$$X^T = \begin{bmatrix} 5 & 3.3 & 1.4 & 0.2 \end{bmatrix}_{1 \times 4 \text{ i.e. } 1 \times d}$$

Let $d: X \times X \rightarrow [0, \infty)$

d be a metric if:

- ① $d(x, y) = d(y, x) \quad \forall x, y \in X$
- ② $d(x, x) = 0 \Rightarrow$ if $d(x, y) = 0$ then $x = y$
- ③ $d(x, y) + d(y, z) \geq d(x, z) \quad \forall x, y, z \in X$

each x_i is D-Dimensional

$$X = \{x_1, x_2, \dots, x_n\}$$

Distance $X \subseteq \mathbb{R}^D$ set of all D-Dimensional real sets

Sample i $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}$ Sample j $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_D \end{bmatrix}$

Distance

Distance be a metric and can be defined as:

$$d_p(x, y) = \left(\sum_{i=1}^D |x_i - y_i|^p \right)^{1/p} \quad ; p \geq 1$$

where $x, y \in X$, $X \subseteq \mathbb{R}^D$

$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_D \end{bmatrix}$

$\| \cdot \|_p$ -norm $\rightarrow d_p(x, y) = \left(\sum_{i=1}^D |x_i - y_i|^p \right)^{1/p}$

$p=1 \rightarrow d_1(x, y) = \sum_{i=1}^D |x_i - y_i|$

d_1 -Norm $\rightarrow$ City Block / Manhattan Distance

$p=2 \rightarrow d_2(x, y) = \left(\sum_{i=1}^D |x_i - y_i|^2 \right)^{1/2}$

d_2 -Norm $\rightarrow$ Euclidean Distance

Euclidean Dist

represents straight dist b/w two points

Euclidean Dist $\rightarrow d(p, q) = \left((p_1 - q_1)^2 + (p_2 - q_2)^2 \right)^{1/2}$
 for 2D

We can generalize for n-Dimensional space as:

$$d_n = \left[\sum_{i=1}^n (p_i - q_i)^2 \right]^{1/2}$$

$n = \text{no of Dimensions}$

Manhattan Dist

Sum of Absolute Differences b/w points across all dimensions

The dist b/w two points measured along axes at right Angle.

For 2D:

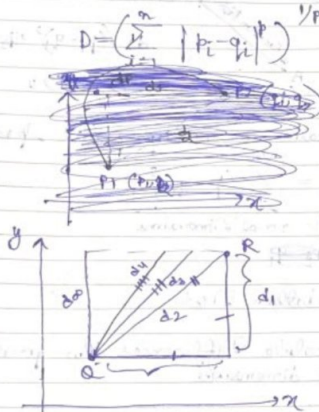
Manhattan Dist $\rightarrow d(p, q) = |p_1 - q_1| + |p_2 - q_2|$

Generalized

for n-Dimensional $d_n = \sum_{i=1}^n |p_i - q_i|$

Minkowski Dist

Generalized form of Euclidean & Manhattan Dist



$$d_1 > d_2 > d_3 > \dots > d_\infty$$

$p = \infty$

$$d_\infty(x, y) = \max_{i=1,2,3,\dots,D} |x_i - y_i| \quad \left(\begin{array}{l} \text{L}_\infty \text{ Norm} \\ \text{or} \\ \text{L/Infinite} \end{array} \right)$$

Restrict to $p=2$

$$d_2(x, y) = \left[\sum_{i=1}^n (x_i - y_i)^2 \right]^{1/2}$$

$$d_2^2(x, y) = (x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2$$

$$= (x_1 - y_1)(x_1 - y_1) + (x_2 - y_2)(x_2 - y_2) + \dots + (x_n - y_n)(x_n - y_n)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$d_2^2(x, y) = \begin{bmatrix} (x_1 - y_1) & (x_2 - y_2) & \dots & (x_n - y_n) \end{bmatrix} \begin{bmatrix} (x_1 - y_1) \\ (x_2 - y_2) \\ \vdots \\ (x_n - y_n) \end{bmatrix}$$

$$Z = \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}_{n \times 1}$$

$$d_2^2(x, y) = Z^T Z = \begin{bmatrix} 1 \times n & n \times n & n \times 1 \end{bmatrix}$$

$$d_2^2(x, y) = Z^T \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}_{n \times n} \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}$$

$$d_2^2(x, y) = Z^T Z$$

$$d_2^2(x, y) = Z^T I Z \geq 0 \text{ always}$$

$d_2^2 = Z^T A Z$; A is Symmetric, Definite matrix
Type of Square matrix i.e. both Symmetric & +ve or -ve definite

Symmetric Matrix

A is symmetric if $A = A^T$

Entries are mirrored across the Diagonal

Definite Matrix: +ve Definite Matrix

All Eigenvalues of A are +ve

For any non-zero vector x , $x^T A x > 0$

-ve Definite Matrix

All Eigen Values of A are -ve

For any non-zero vector x , $x^T A x < 0$

Positive Symmetric Matrix, Prop

- ① $x^T A x > 0$
- ② $|A| > 0$
- ③ A^{-1} is also +ve Symmetric
- ④ All Eigen Values of $A > 0$

Symmetric +ve definite matrix
 $x^T A x > 0$ for any vector x

$$d_2^2 = Z^T A_{n \times n} Z$$

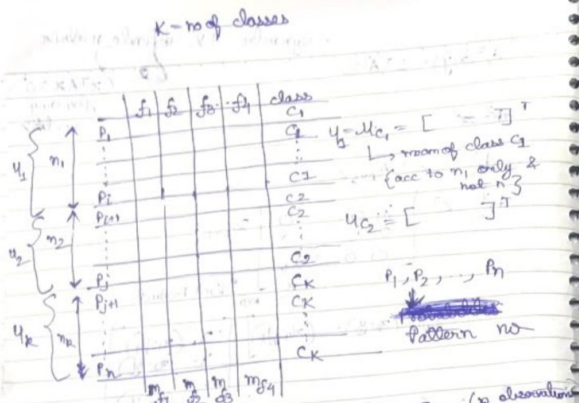
$$d_2^2 = Z^T \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} Z_{n \times 1}$$

$$= \begin{bmatrix} (x_1 - y_1) & (x_2 - y_2) & \dots & (x_n - y_n) \end{bmatrix} \begin{bmatrix} \lambda_1 (x_1 - y_1) \\ \lambda_2 (x_2 - y_2) \\ \vdots \\ \lambda_n (x_n - y_n) \end{bmatrix}$$

$$d_2^2 = \lambda_1 (x_1 - y_1)^2 + \lambda_2 (x_2 - y_2)^2 + \dots + \lambda_n (x_n - y_n)^2$$

Sum of Weighted Euclidean Distances

$$= \sum_{i=1}^n \lambda_i |x_i - y_i|^2$$



Let a be a variable/feature $\rightarrow a_1, a_2, \dots, a_n$ (no observation)

$$m_a = \frac{1}{n} \sum_{i=1}^n a_i$$

$$\text{Mean of Data} \quad \underline{u} = [m_{a_1} \ m_{a_2} \ \dots \ m_{a_n}]^T$$

$$\underline{u} = \begin{bmatrix} m_{a_1} \\ m_{a_2} \\ \vdots \\ m_{a_n} \end{bmatrix}$$

$$\underline{u} = \begin{bmatrix} m_{a_1} \\ m_{a_2} \\ \vdots \\ m_{a_n} \end{bmatrix}$$

$$\underline{u}_1 = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \end{bmatrix}$$

$$u_{11} = \frac{1}{n_1} (f_{11} + f_{12} + \dots + f_{1n})$$

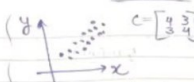
$$u_{12} = \frac{1}{n_1} (f_{21} + f_{22} + \dots + f_{2n})$$

$$u_{13} = \frac{1}{n_1} (f_{31} + f_{32} + \dots + f_{3n})$$

$$\vdots$$

$$u_{1n} = \frac{1}{n_1} (f_{n1} + f_{n2} + \dots + f_{nn})$$

Note: +ve Covariance



$$\text{Cov}(f_i, f_i) = \text{Var}(f_i)$$

$$\text{Cov}(f_i, f_j) = \text{Cov}(f_j, f_i)$$

$$\Sigma = \begin{bmatrix} \text{Var}(f_1) & \text{Cov}(f_1, f_2) & \dots & \text{Cov}(f_1, f_n) \\ \text{Cov}(f_2, f_1) & \text{Var}(f_2) & \dots & \text{Cov}(f_2, f_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_n, f_1) & \text{Cov}(f_n, f_2) & \dots & \text{Var}(f_n) \end{bmatrix}_{n \times n}$$

	f_1	f_2	f_3
1	1	10	5
2	2	15	8
3	3	20	12

$$\underline{u} = \begin{bmatrix} 2 \\ 15 \\ 5 \end{bmatrix}$$

$$\text{Var}(f_1) = \frac{1}{3} \{ (1-2)^2 + (2-2)^2 + (3-2)^2 \} = \frac{2}{3}$$

$$\text{Var}(f_2) = \frac{1}{3} \{ (-5)^2 + 0^2 + (5)^2 \} = \frac{50}{3}$$

$$\text{Var}(f_3) = \frac{1}{3} \{ 0^2 + 8^2 + (5-3)^2 \} = \frac{18}{3} = 6$$

$$\text{Cov}(f_1, f_2) = \frac{1}{3} [-1 \times 5 + 0 \times 0 + 1 \times 5] = \frac{0}{3} = 0$$

Variance

$$V = \frac{1}{n} \sum_{i=1}^n (a_i - m_a)^2 = \frac{1}{n} \sum_{i=1}^n (a_i - m_a) (a_i - m_a)$$

$$\text{Cov}(a, b) = \frac{1}{n} \sum_{i=1}^n (a_i - m_a) (b_i - m_b)$$

if we change a , how b changes $\rightarrow$ Vice Versa



$$\text{Cov}(a, b) > 0$$

if $a \uparrow \Rightarrow b \uparrow$
if $b \uparrow \Rightarrow a \uparrow$

$$\text{Cov}(a, b) < 0$$

if $a \uparrow \Rightarrow b \downarrow$
if $b \uparrow \Rightarrow a \downarrow$

$$\text{Cov}(a, b) = 0$$

a & b are independent

Shows the Covariance b/w each pair of elements in a random vector

Dispersion Matrix / Covariance Matrix / Variance-Covariance Matrix

$$\Sigma = \begin{bmatrix} \text{Cov}(f_1, f_1) & \text{Cov}(f_1, f_2) & \dots & \text{Cov}(f_1, f_n) \\ \text{Cov}(f_2, f_1) & \text{Cov}(f_2, f_2) & \dots & \text{Cov}(f_2, f_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_n, f_1) & \text{Cov}(f_n, f_2) & \dots & \text{Cov}(f_n, f_n) \end{bmatrix}_{n \times n}$$

$$\text{Cov}(f_1, f_3) = \frac{1}{3} [-1 \times 0 + 0 \times 3 + 1 \times 5] = \frac{5}{3} = 1.67$$

$$\text{Cov}(f_2, f_3) = \frac{1}{3} [-5 \times 0 + 0 \times 3 + 5 \times 5] = \frac{25}{3} = 8.33$$

$$\Sigma = \begin{bmatrix} 2/3 & 0 & 5/3 \\ 0 & 50/3 & 25/3 \\ 5/3 & 25/3 & 6 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 0 & 5 \\ 0 & 50 & 25 \\ 5 & 25 & 18 \end{bmatrix}$$

$$A = \begin{bmatrix} f_1 & f_2 & f_3 \\ f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} = \begin{bmatrix} 1 & 10 & 5 \\ 2 & 15 & 8 \\ 3 & 20 & 12 \end{bmatrix}$$

$$\underline{u} = \begin{bmatrix} 2 \\ 15 \\ 5 \end{bmatrix}$$

Mean Subtracted Dataset

$$A = B = \begin{bmatrix} f_1 - u_1 & f_2 - u_2 & f_3 - u_3 \\ f_{11} - u_{11} & f_{12} - u_{12} & f_{13} - u_{13} \\ f_{21} - u_{21} & f_{22} - u_{22} & f_{23} - u_{23} \\ f_{31} - u_{31} & f_{32} - u_{32} & f_{33} - u_{33} \end{bmatrix} = \begin{bmatrix} -1 & -2 & -3 \\ 0 & 5 & 3 \\ 1 & 5 & 7 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 0 & 1 \\ -5 & 0 & 5 \\ 0 & 3 & -3 \end{bmatrix}$$

BBT matrix

$$B = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} = \begin{bmatrix} -4 & -4 & -4 \\ -4 & -4 & -4 \\ -4 & -4 & -4 \end{bmatrix}$$

$$B^T = \begin{bmatrix} -4 & -4 & -4 \\ -4 & -4 & -4 \\ -4 & -4 & -4 \end{bmatrix}$$

$$BB^T = \begin{bmatrix} -1 & 0 & 1 \\ -5 & 0 & 5 \\ 0 & 3 & -3 \end{bmatrix}$$

But we didn't get $1/3$

$Z = \frac{1}{\sqrt{3}} BBT = \frac{1}{\sqrt{3}} BBT$

Where B = mean Subtracted Matrix

doesn't affect the conclusion of Covariance

$$Z = BB^T = B^T B$$

Z is Symmetric +ve Definite Matrix

$Z \rightarrow$ Real λ
 $Z \rightarrow$ Symmetric $\lambda \Rightarrow Z^{-1} \rightarrow$ +ve Definite
 Positive Definite $\lambda \Rightarrow \lambda^T A \lambda > 0$

$$\lambda^T Z \lambda > 0$$

Verify:

$$\begin{bmatrix} -1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -10 & -8 \\ 10 & 50 & -15 \\ -3 & -15 & 8 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 12 & -100 & -64 \\ 10 & 50 & -15 \\ -3 & -15 & 8 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix}$$

$$= -12 + 130 \times 3 - 2 \times 64$$

$$= 570 - 12 - 128$$

$$= 540 - 140$$

$$= 400 > 0$$

n features N obsⁿ

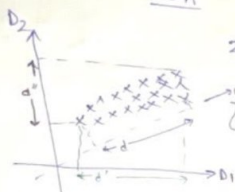
$$F_{\text{New}} = \begin{bmatrix} e_1^T & e_1^T & \dots & e_1^T \\ e_2^T & e_2^T & \dots & e_2^T \\ \vdots & \vdots & \ddots & \vdots \\ e_p^T & e_p^T & \dots & e_p^T \end{bmatrix} \begin{bmatrix} f_{11} - \bar{f}_1 & f_{12} - \bar{f}_1 & \dots & f_{1N} - \bar{f}_1 \\ f_{21} - \bar{f}_2 & f_{22} - \bar{f}_2 & \dots & f_{2N} - \bar{f}_2 \\ \vdots & \vdots & \ddots & \vdots \\ f_{p1} - \bar{f}_p & f_{p2} - \bar{f}_p & \dots & f_{pN} - \bar{f}_p \end{bmatrix}$$

1st feature
2nd feature
nth feature
1st pattern
2nd pattern
nth pattern

$$\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_p \end{bmatrix} \begin{bmatrix} e_1^T (f_{11} - \bar{f}_1) + \dots + e_1^T (f_{1N} - \bar{f}_1) & e_1^T (f_{12} - \bar{f}_2) + \dots + e_1^T (f_{1N} - \bar{f}_2) & \dots \\ e_2^T (f_{21} - \bar{f}_1) + \dots + e_2^T (f_{2N} - \bar{f}_1) & e_2^T (f_{22} - \bar{f}_2) + \dots + e_2^T (f_{2N} - \bar{f}_2) & \dots \\ \vdots & \vdots & \ddots & \vdots \\ e_p^T (f_{p1} - \bar{f}_1) + \dots + e_p^T (f_{pN} - \bar{f}_1) & e_p^T (f_{p2} - \bar{f}_2) + \dots + e_p^T (f_{pN} - \bar{f}_2) & \dots \end{bmatrix}$$

1st new feature
pth new feature
1st pattern
2nd pattern
nth pattern

PCA



2D → 1D

→ I need to find a new axis on which I can project my data & get similar results for what I was getting in 2D

I need to find Unit Vector

Single pt Eg



I want to project this vector onto another vector

Projection of $\vec{x}$ along $\vec{u} = \frac{\vec{x} \cdot \vec{u}}{|\vec{u}|} \vec{u}$

$$= \vec{x} \cdot \vec{u} = u^T x$$

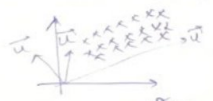
$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = x_1 u_1 + x_2 u_2$$

↑
Scalar (gives dist)

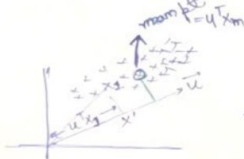
Similarly, there will be many such pts



Goal: I want to maximize the variance



there can be many $\vec{u}$ possible, but we want that vector for which we get max variance



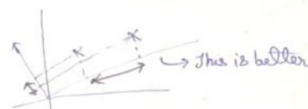
$$[u^T x_1] \quad [u^T x_2] \quad \dots \quad [u^T x_n]$$

→ After projecting all the pts, I will cal variance

$$\text{Var} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\frac{1}{n} \sum_{i=1}^n (u^T x_i - u^T \bar{x})^2 = \text{variance} \rightarrow \text{mathematical objective fn}$$

Mathematically, Using PCA → I've to find a unit vector $\vec{u}$ for which variance is maximum



Covariance matrix

$$\Sigma_{D \times D} = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

$$\text{Cov}(x_i, x_i) = \text{Var}(x_i)$$

$$\text{Cov}(x_i, x_j) = \text{Cov}(x_j, x_i)$$

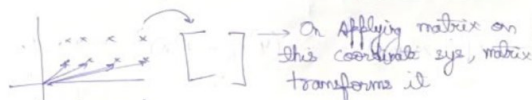
→ special matrix: Tells the spread of all your features (Every Axis has its variance) + Tells relationship b/w two variables

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} V_x & C_{xy} & C_{xz} \\ C_{yx} & V_y & C_{yz} \\ C_{zx} & C_{zy} & V_z \end{bmatrix}$$

Eigen Decomposition of Covariance Matrix

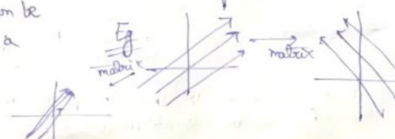
↓
Finding Eigen Value & Eigen Vector

Linear Transformation



Each pt can be treated as a vector

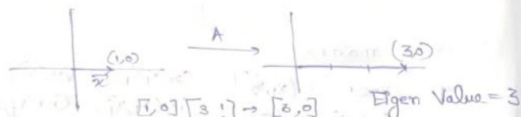
$$\vec{x} \rightarrow A\vec{x}$$



Eigen Vectors: Sp Vectors, if you apply matrix on them, their dirⁿ don't change (Magnitude may change) (or stretch)

Eigen Values: The amount of stretch in eigen vector on applying linear transformation

Ex $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$



$\rightarrow \vec{x}$ is an Eigen Vector
 $\rightarrow \lambda = 3$

Eigen Vector property

$A\vec{v} = \lambda\vec{v}$
 Matrix Eigen Vector Eigen Value (Scalar)

Why PCA uses Eigen Decomposition of a Covariance matrix

Covariance matrix defines both the spread (variance) & orientation (covariance) of our data

optimization $f = \sum \frac{(y^T x_i - y^T \bar{x})^2}{n}$ Find y s.t. var becomes max

Ex $D = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 4 & 5 \end{bmatrix} \rightarrow$ Data

If we define vector $\vec{v}$ then the projection of our data on this vector is obtained $\vec{v}^T D$.

Variance of projected data is $\vec{v}^T Z \vec{v}$.

We're looking for $\vec{v}$ that points into the dirⁿ of largest variance, we should choose $\vec{v}$ s.t. $\vec{v}^T Z \vec{v}$ is large as possible ($\vec{v}$ is unit vector)

$\rightarrow$ Solved by Rayleigh Quotient (Technique)

$\rightarrow$ It was found, the largest Eigen vector of the covariance matrix always points into the dirⁿ of largest variance of the data.
 Largest Eigen Vector $\Rightarrow$ Largest Eigen Value

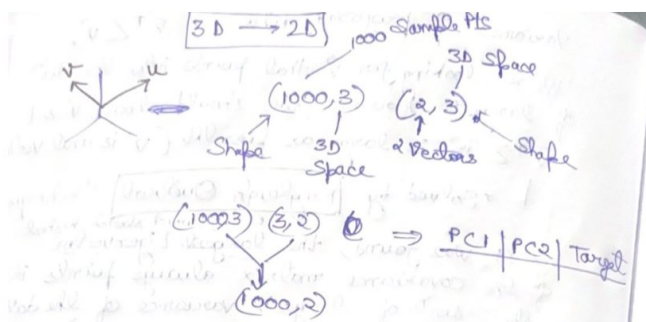
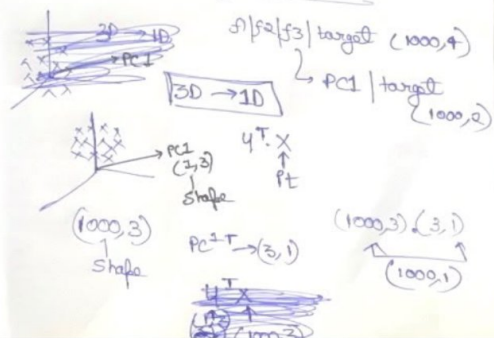
Optimization f is solved by technique (Rayleigh Quotient)

$\therefore$ Do Eigen Decomposition of Covariance Matrix

Find Eigen Values & Eigen Vectors

Find Largest Eigen Vector

How to transform Points



Principal Component Analysis (PCA)

→ Method of feature Extraction
(High Dimensional Data → Low Dimensional Data)

~~Step 1:~~

Given data in Table, dimension = no. of features

To Cal Eigen Values & Eigen Vectors of Matrix A

$$A_{D \times D} \vec{x} = \lambda \vec{x}$$

$$(A - \lambda I) \vec{x} = 0$$

$|A - \lambda I| = 0 \rightarrow$ 'D' no. of eigen values $\lambda_1, \lambda_2, \dots, \lambda_D$

'D' no. of Eigen vectors $V_1, V_2, \dots, V_D$

$$\Sigma_{D \times D} = \begin{bmatrix} \text{Var}(f_1) & \text{Cov}(f_1, f_2) & \dots & \text{Cov}(f_1, f_D) \\ \text{Cov}(f_2, f_1) & \text{Var}(f_2) & \dots & \text{Cov}(f_2, f_D) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_D, f_1) & \dots & \dots & \text{Var}(f_D) \end{bmatrix}$$

→ Symmetric +ve Definite Matrix
 $\Sigma_{D \times D} = A_{D \times D}$

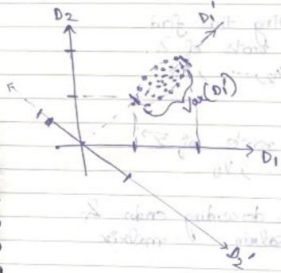
(i) $\lambda_i > 0 \forall i = 1, 2, \dots, D$ (All Eigen values are +ve)

$$(ii) |\Sigma_{D \times D}| = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_D = \prod_{i=1}^D \lambda_i$$

$$(iii) \text{Tr}(\Sigma) = \text{Trace} = \text{Sum of Elements on Main Diagonal} = \lambda_1 + \lambda_2 + \dots + \lambda_D = \sum_{i=1}^D \lambda_i = \sum_{i=1}^D \text{Var}(f_i)$$

$$(iv) V_i \cdot V_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

→ All Eigen vectors are $\perp$ to each other



We can't discriminate Data on basis of D_2 but can highly discriminate Data on basis of D_1

Steps of PCA:

- (i) Get the data
- (ii) Perform mean subtraction over the data

$$\Sigma_{D \times D} = \frac{1}{N} A_{\text{meas}} (A_{\text{meas}})^T$$

$$\Sigma_{D \times D} = \begin{bmatrix} \text{Var}(f_1) & \text{Cov}(f_1, f_2) & \dots & \text{Cov}(f_1, f_D) \\ \text{Cov}(f_2, f_1) & \text{Var}(f_2) & \dots & \text{Cov}(f_2, f_D) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_D, f_1) & \text{Cov}(f_D, f_2) & \dots & \text{Var}(f_D) \end{bmatrix}$$

- (iii) Cal Eigen Values

$$|Z - \lambda I| = 0 \quad \text{solving this find } D \text{ roots of } \lambda.$$

- (iv) Cal Eigen ~~Vectors~~ $\lambda_1, \lambda_2, \dots, \lambda_D$

$$(Z - \lambda I) \vec{x} = 0$$

Solving this find D roots of $\vec{x}$,
 $v_1, v_2, \dots, v_D$

- (v) Sort the Eigen values in descending order & form the transformation matrix

Ex $D=2$ (Two features) $\lambda_1 > \lambda_2$

$$W = [v_1 \ v_2]$$

$$\lambda_1 > \lambda_2$$

$$W = [v_1 \ v_2]$$

- (vi) ~~Transform~~ Transform the data using $y = W^T x$

Ex $D=2$

$D_1 \rightarrow 1^{\text{st}}$ feature $D_2 \rightarrow 2^{\text{nd}}$ feature

data in feature space

$$\begin{bmatrix} \text{Sample} \\ y \\ \vdots \end{bmatrix} D' = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \text{Sample} \\ x \\ \vdots \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}$$

Delete the Columns from last of W to reduce the value of D to d .

If $\sum_{i=1}^d \lambda_i = 0.95$ or 0.98 $\rightarrow$ We select 95% of Variance

Note: For $A \in \mathbb{R}^{n \times n}$ n -Dimension
 n -features
 Eigen values $\rightarrow \lambda_1, \lambda_2, \dots, \lambda_n$

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be Order

Corresponding unit eigen vectors be $e_1, e_2, \dots, e_n$

→ choose a +ve integer p s.t. $1 \leq p \leq n$

If $p=1$

To Convert n -D to 1-D, $p=1$

To Convert n -D to 2-D, $p=2$

To Convert n -D to i -D, $p=i$

To Convert n -D to n -D, $p=n$

Ex To change n -D to 5-D
 i.e. Selecting 5 features
 $p=5$

→ Select top p no of Eigen Values
 & top p Eigen Vectors

$$F = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_p^T \end{bmatrix} \text{ not } 1^{\text{st}} \text{ to } n$$

To Derive new Dataset

$$A = \begin{bmatrix} f_{11} - \bar{f}_1 & f_{12} - \bar{f}_2 & \dots & f_{1N} - \bar{f}_N \\ f_{21} - \bar{f}_1 & f_{22} - \bar{f}_2 & \dots & f_{2N} - \bar{f}_N \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} - \bar{f}_1 & f_{n2} - \bar{f}_2 & \dots & f_{nN} - \bar{f}_N \end{bmatrix} n \times N$$

$$f = \begin{bmatrix} f_{11} & f_{12} & \dots & f_{1N} \\ f_{21} & f_{22} & \dots & f_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \dots & f_{nN} \end{bmatrix} n \times N$$

$$F = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_p^T \end{bmatrix} p \times 1$$

$$\text{make } F_{p \times N} = \begin{bmatrix} e_1^T & e_1^T & \dots & e_1^T \\ e_2^T & e_2^T & \dots & e_2^T \\ \vdots & \vdots & \ddots & \vdots \\ e_p^T & e_p^T & \dots & e_p^T \end{bmatrix} p \times N$$

$$F_{\text{new}} = F \cdot A$$

dataset of N samples having p features

$$F_{\text{new}} = \begin{bmatrix} f_{11} - \bar{f}_1 & f_{12} - \bar{f}_2 & \dots & f_{1N} - \bar{f}_N \\ f_{21} - \bar{f}_1 & f_{22} - \bar{f}_2 & \dots & f_{2N} - \bar{f}_N \\ \vdots & \vdots & \ddots & \vdots \\ f_{p1} - \bar{f}_1 & f_{p2} - \bar{f}_2 & \dots & f_{pN} - \bar{f}_N \end{bmatrix}$$

$$F_{\text{New}} = \begin{bmatrix} e_1^T & e_1^T & \dots & e_1^T \\ e_2^T & e_2^T & \dots & e_2^T \\ \vdots & \vdots & \ddots & \vdots \\ e_p^T & e_p^T & \dots & e_p^T \end{bmatrix} \begin{bmatrix} f_{11} - \bar{f}_1 & f_{12} - \bar{f}_2 & \dots & f_{1N} - \bar{f}_N \\ f_{21} - \bar{f}_1 & f_{22} - \bar{f}_2 & \dots & f_{2N} - \bar{f}_N \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} - \bar{f}_1 & f_{n2} - \bar{f}_2 & \dots & f_{nN} - \bar{f}_N \end{bmatrix}$$

$p \times n$ 1st Pattern 2nd Pattern n^{th} Pattern

$$= \begin{bmatrix} f_1' & f_2' & \vdots & f_p' \end{bmatrix} \begin{bmatrix} e_1^T(f_{11} - \bar{f}_1) + \dots + e_1^T(f_{n1} - \bar{f}_n) & e_1^T(f_{12} - \bar{f}_2) + \dots + e_1^T(f_{n2} - \bar{f}_n) & \dots & e_1^T(f_{1N} - \bar{f}_N) + \dots + e_1^T(f_{nN} - \bar{f}_n) \\ e_2^T(f_{11} - \bar{f}_1) + \dots + e_2^T(f_{n1} - \bar{f}_n) & e_2^T(f_{12} - \bar{f}_2) + \dots + e_2^T(f_{n2} - \bar{f}_n) & \dots & e_2^T(f_{1N} - \bar{f}_N) + \dots + e_2^T(f_{nN} - \bar{f}_n) \\ \vdots & \vdots & \ddots & \vdots \\ e_p^T(f_{11} - \bar{f}_1) + \dots + e_p^T(f_{n1} - \bar{f}_n) & e_p^T(f_{12} - \bar{f}_2) + \dots + e_p^T(f_{n2} - \bar{f}_n) & \dots & e_p^T(f_{1N} - \bar{f}_N) + \dots + e_p^T(f_{nN} - \bar{f}_n) \end{bmatrix}$$

1st new feature 2nd new feature p^{th} new feature

1st Pattern 2nd Pattern n^{th} Pattern

Ex1 Given data in Table, Reduce the dimension from 2 to 1 using PCA Algo

	f_1	f_2
P1	4	11
P2	8	4
P3	13	5
P4	7	14

S1:

$$\mu_{f_1} = \frac{4+8+13+7}{4} = 8$$

$$\mu_{f_2} = \frac{11+4+5+14}{4} = 8.5$$

$$\text{Cov}(f_1, f_2) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)$$

$$= \frac{1}{4} [(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)]$$

$$= \frac{1}{4} [-10 + 0 - 17.5 - 5.5] = -11$$

For Sample, $\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

For pop, $\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$

$$\text{Var}(f_1) = \frac{(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2}{4}$$

$$\text{Var}(f_2) = \frac{(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2}{4}$$

$$Z_{\text{axe}} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

S2: Cal Eigen Values of Z_{axe}

$$|\Sigma - \lambda I| = 0$$

$$\begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (14-\lambda)(23-\lambda) - (121) = 0$$

$$\Rightarrow \lambda^2 - 37\lambda + 201 = 0$$

$$\lambda = \frac{37 \pm \sqrt{(37)^2 - 4(201)}}{2}$$

$$\lambda = 30.3849, 6.6151$$

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

S3: $\lambda_1 = 30.3849$
 $\lambda_2 = 6.6151$

Cal Eigen Vectors

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = (\Sigma - \lambda I) \vec{v}$$

$$= \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} (14-\lambda)v_1 - 11v_2 \\ -11v_1 + (23-\lambda)v_2 \end{bmatrix}$$

$$(14-\lambda)v_1 - 11v_2 = 0 \quad \text{--- (1)}$$

$$-11v_1 + (23-\lambda)v_2 = 0 \quad \text{--- (2)}$$

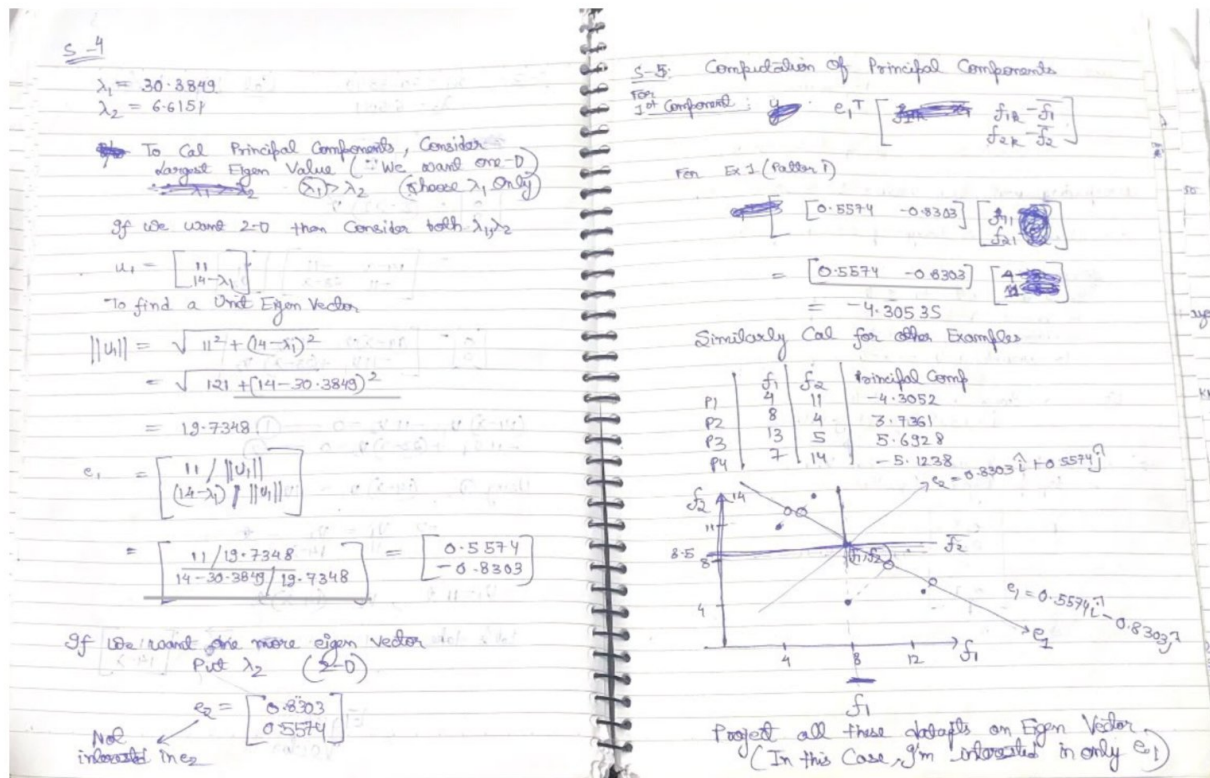
Using (1), $(14-\lambda)v_1 = 11v_2$

$$\Rightarrow \frac{v_1}{v_2} = \frac{11}{14-\lambda}$$

$$v_1 = 11t, v_2 = (14-\lambda)t$$

Let's take $t=1$ $\vec{v} = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}$

Eigen Vector



Properties:

1. **Sum of Eigenvalues:** The sum of all the eigenvalues of a matrix is equal to its trace (the sum of the diagonal elements of the matrix). This holds true regardless of whether the matrix is square or not.
2. **Product of Eigenvalues:** The product of all the eigenvalues of a matrix is equal to its determinant. This also holds for square matrices.
3. **Eigenvectors corresponding to different eigenvalues are orthogonal:** If a matrix A is symmetric (i.e., $A = A^T$), the eigenvectors corresponding to distinct eigenvalues are orthogonal to each other.
4. **Eigenvalue of a Identity Matrix:** For an identity matrix, the eigenvalues are all 1, regardless of the dimension of the matrix.
5. **Eigenvalue of a Scalar Multiple:** If B is a matrix obtained by multiplying a scalar c to a matrix A (i.e., $B = cA$), then the eigenvalues of B are just the eigenvalues of A each multiplied by c.
6. **Eigenvalues of a Diagonal Matrix:** For a diagonal matrix, the eigenvalues are the diagonal elements themselves.

Diagonal Matrix

$$A = \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{33} & 0 & 0 \\ 0 & 0 & 0 & a_{44} & 0 \\ 0 & 0 & 0 & 0 & a_{55} \end{bmatrix}_{5 \times 5}$$

7. Eigenvalues of a Transposed Matrix: The eigenvalues of a matrix and its transpose are the same.

PCA Variants

Kernal PCA

Incremental PCA

Randomized PCA

Sp Matrices:

Diagonal Matrix:

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & 0 \\ 0 & 0 & 0 & a_{44} \end{bmatrix}_{D \times D}$$

→ Entries outside Diagonal are all 0

Properties:

$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \Rightarrow A^{100} = \underbrace{A \cdot A \cdot \dots \cdot A}_{100 \text{ times}} = \begin{bmatrix} a^{100} & 0 \\ 0 & b^{100} \end{bmatrix}$

n^{th} power of Diag matrix can be obtained by raising each diagonal element to the power of n

Eigen Values:

$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \rightarrow \lambda_1 = a, \lambda_2 = b$

Matrix Multiplication by Vector

It scales each component of vector by the corresponding element on the diagonal

Matrix Multiplication

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix}$$

Orthogonal matrix

→ Sq matrix whose columns & rows are orthogonal unit vectors (i.e orthogonal vectors)

→ All are of unit length & are at right angles to each other

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is Orthogonal if

- $\vec{v}_1 = \begin{bmatrix} a \\ c \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$ i.e $\vec{v}_1 \cdot \vec{v}_2 = 0$
 $ab + cd = 0$
- $\vec{v}_1 \cdot \vec{v}_1 = 1, \vec{v}_2 \cdot \vec{v}_2 = 1$
 $a^2 + c^2 = 1, b^2 + d^2 = 1$
 $ac + bd = 0$
- $\sqrt{a^2 + c^2} = 1, \sqrt{b^2 + d^2} = 1$
- $\sqrt{a^2 + b^2} = 1, \sqrt{c^2 + d^2} = 1$

Note: Any matrix of the form: $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ is orthogonal for any θ .

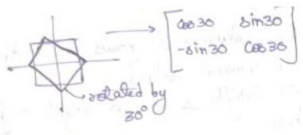
$\begin{bmatrix} \cos 30 & \sin 30 \\ -\sin 30 & \cos 30 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

- $\frac{\sqrt{3}}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 0$
- $\frac{\sqrt{3}}{2} \cdot (-\frac{1}{2}) + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 0$
- $\sqrt{(\frac{\sqrt{3}}{2})^2 + (-\frac{1}{2})^2} = 1, \sqrt{(\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = 1$
- $\sqrt{(\frac{\sqrt{3}}{2})^2 + (\frac{1}{2})^2} = 1, \sqrt{(-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = 1$

Note: Identity matrix is also Orthogonal

Properties:

- Inverse: $A^T = A^{-1}$
- On linear transformation, it gives perfect rotation, no scaling

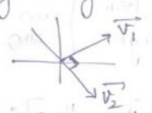


3) Symmetric Matrix
 → sq matrix s.t. $A^T = A$

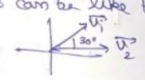
$$\begin{bmatrix} a & c \\ c & b \end{bmatrix}^T = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$$

Properties:

- A) Real Eigenvalues: $\lambda \in \mathbb{R}$ (not Complex)
- B) Orthogonal Eigen vectors: $\vec{v}_1 \perp \vec{v}_2$



for other matrices, Eigen vectors can be like this



Note: Covariance matrix

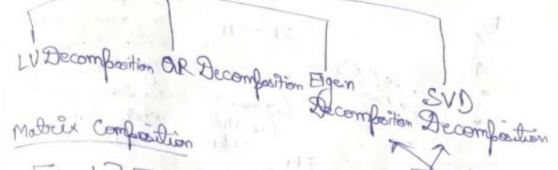
$$\text{gpa} | \text{iq} \quad \Sigma = \begin{bmatrix} \text{var}(\text{gpa}) & \text{Cov}(\text{gpa}, \text{iq}) \\ \text{Cov}(\text{gpa}, \text{iq}) & \text{var}(\text{iq}) \end{bmatrix}$$

~~Covariance matrix~~

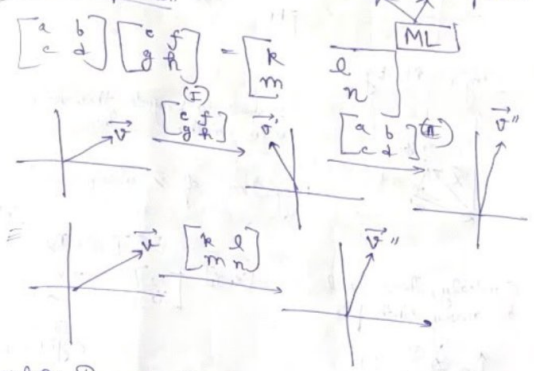
Symmetric matrix $\Rightarrow$ Eigen Vectors are $\perp$ to each other

Note: orthogonal eigen vectors $\Rightarrow$ Symmetric matrix

Matrix Decomposition



Matrix Composition



Matrix Decomposition

$$ABC = D \quad (\text{Composition})$$

$$D = ABC \quad (\text{Decomposition})$$

$$A = BC^T$$

• First C is applied then B

$$\vec{v} \xrightarrow{[C]} \vec{v'} \xrightarrow{[B]} \vec{v''} \text{ is applied.}$$

$$\vec{v} \xrightarrow{[A]} \vec{v''}$$

Eigen Decomposition

Eigen Decomposition of matrix A is given by Eqⁿ:

$$A = V \Lambda V^{-1} \quad V, \Lambda \rightarrow \text{Matrices}$$

$V \rightarrow$ Matrix whose columns are eigenvectors of A

$\Lambda \rightarrow$ Diagonal matrix whose entries are eigenvalues of A

$V^{-1} \rightarrow$ Inverse of V

Assumptions of A

1. Square Matrix

$\downarrow$
Eigen Decomposition defined for square matrices only

2. $\bullet$ Diagonalizability

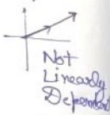
$\downarrow$
For $n \times n$ matrix, it should have n linearly independent eigenvectors

Linearly Independent Vectors

Set $\{v_1, v_2, \dots, v_n\}$ is linearly independent if:

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$$

$\rightarrow$ Only when all scalars $c_i = 0$


Not Linearly Dependent

Geometric meaning:

Two vectors are independent if they are not parallel (collinear). Each vector adds a new dimension to the space.

Eg $\{ (1,0), (0,1) \}$ in $\mathbb{R}^2$

$\{ (1,0,0), (0,1,0), (0,0,1) \}$ in $\mathbb{R}^3$

$\rightarrow$ Orthogonal vectors are always linearly independent.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \xrightarrow{\lambda_1, \lambda_2} \vec{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

Eigen Vectors

(Eigen Decomposition Theorem)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}^{-1}$$

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 = 0 \quad \text{only when } c_1 = c_2 = 0$$

Eigen Vector: $A \vec{v} = \lambda \vec{v} \Rightarrow \begin{matrix} A \vec{v}_1 = \lambda_1 \vec{v}_1 \\ A \vec{v}_2 = \lambda_2 \vec{v}_2 \end{matrix}$

$$\begin{matrix} A \vec{v}_1 = \lambda_1 \vec{v}_1 \\ A \vec{v}_2 = \lambda_2 \vec{v}_2 \end{matrix} \Rightarrow \boxed{A V = V \Lambda}$$

Matrix Form

$$V = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$A \vec{v}_1 = \lambda_1 \vec{v}_1 \quad \vec{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}_{2 \times 1} = \lambda_1 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}_{2 \times 1} \Rightarrow \begin{bmatrix} ax_1 + by_1 \\ cx_1 + dy_1 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 \\ \lambda_1 y_1 \end{bmatrix}$$

Eg: $\begin{matrix} ax_1 + by_1 = \lambda_1 x_1 & \text{--- (1)} \\ cx_1 + dy_1 = \lambda_1 y_1 & \text{--- (2)} \end{matrix}$

$$A \vec{v}_2 = \lambda_2 \vec{v}_2 \Rightarrow \begin{matrix} ax_2 + by_2 = \lambda_2 x_2 & \text{--- (3)} \\ cx_2 + dy_2 = \lambda_2 y_2 & \text{--- (4)} \end{matrix}$$

$$A V = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} ax_1 + by_1 & ax_2 + by_2 \\ cx_1 + dy_1 & cx_2 + dy_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 \\ \lambda_1 y_1 & \lambda_2 y_2 \end{bmatrix}$$

$$\Rightarrow \begin{matrix} ax_1 + by_1 = \lambda_1 x_1 & \text{--- (1)} \\ cx_1 + dy_1 = \lambda_1 y_1 & \text{--- (2)} \end{matrix}$$

$$AV = VA$$

$$\Rightarrow A = V \Lambda V^{-1} \rightarrow \text{Eigen Decomp}$$

Eigen Vector Diagonal of Eigen Values

Eigen Decomposition of Symmetric Matrix

$$A = V \Lambda V^{-1} \Rightarrow \text{Spectral Decomposition}$$

Symmetric Eigen Vectors of A

$V \rightarrow$ Orthogonal matrix
 $A \rightarrow$ Symmetric matrix
 $\Lambda \rightarrow$ Diagonal matrix
 $V^{-1} \rightarrow$ Orthogonal matrix
 $V \rightarrow$ Rotates
 $\Lambda \rightarrow$ It Scales
 $V^{-1} \rightarrow$ Rotates
 V^{-1} Does rotation in opposite dir

$$A = V \Lambda V^{-1}$$

Rotation Scaling Rotation

1) Rotation from V^{-1}
 2) Scaling from Λ
 3) Rotation from V

Note: PCA assumes that the principal components are a linear combination of the original features

Let two features f_1, f_2
 $PC = a f_1 + b f_2$

It cannot handle complex polynomial relationships b/w features.

Advantages of Eigen Decomposition:

- 1) Used in ML $\rightarrow$ PCA
- 2) Used in Physics
- 3) Used in Signal Theory
- 4) Used in Quantum Theory

$$A_{n \times n} \rightarrow A^{1000}_{n \times n} ?$$

Use Case:

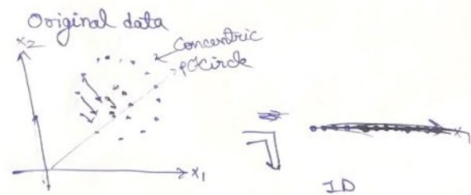
$$A = V \Lambda V^{-1}$$

$$A^{100} = \underbrace{V \Lambda V^{-1} V \Lambda V^{-1} \dots V \Lambda V^{-1}}_{100 \text{ times}} = V \Lambda^{100} V^{-1}$$

2 matrix multiplications $\rightarrow$ Diagonal matrix

Kernel PCA

Kernel PCA $\rightarrow$ PCA + SVM



Data is non-linear $\Rightarrow$ PCA fails $\Rightarrow$ Kernel PCA Works

$\rightarrow$ Kernel PCA uses Kernel Trick

2D Data $\Rightarrow$ 3D Data $\Rightarrow$ 1D Data

